

COBOURG, ON, Canada

WMO: 714310

Lat: 43.9561N

Lon: 78.1511W

Elev: 78

StdP: 100.40

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-16.2	-28.1	0.3	-18.4	-23.8	0.4	-15.1	14.3	-1.9	13.0	-1.5	3.4	320	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	25.9	21.2	24.8	20.9	23.9	20.5	22.8	24.7	22.0	23.9	21.2	23.1	3.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	17.0	24.1	21.3	16.1	23.4	20.4	15.3	22.7	68.0	24.7	64.9	24.0	61.9	23.1	
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Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
11.3	9.4	8.0	-22.7	28.9	3.1	2.1	-24.9	30.4	-26.7	31.7	-28.4	32.9	-30.6	34.4		
			-23.1	24.1	2.8	1.0	-25.1	24.8	-26.8	25.4	-28.4	25.9	-30.4	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-5.1	-4.2	-0.2	5.9	11.4	16.4	19.6	19.7	16.4	9.8	3.8	-1.4
	DBStd	9.70	6.40	5.15	4.85	3.50	3.42	3.01	2.64	2.51	3.35	3.91	4.14	5.05
	HDD10.0	1931	469	397	317	129	24	0	0	0	1	53	186	353
	HDD18.3	4014	727	631	574	374	217	72	17	15	76	266	434	612
	CDD10.0	1107	0	0	1	6	68	193	299	300	193	46	2	0
	CDD18.3	148	0	0	0	0	1	16	57	57	18	0	0	0
	CDH23.3	333	0	0	0	0	3	23	137	137	33	0	0	0
	CDH26.7	23	0	0	0	0	0	1	14	8	0	0	0	0
Wind	WSAvg	3.3	4.2	4.1	3.8	3.6	2.9	2.4	2.2	2.4	2.7	3.2	3.8	4.1
Precipitation	PrecAvg	882	66	53	60	79	77	76	82	76	81	83	72	72
	PrecMax	1039	127	110	124	126	143	155	166	124	154	167	112	147
	PrecMin	753	34	24	28	28	24	19	36	28	45	27	30	42
	PrecStd	89	26	20	24	30	33	38	36	22	27	35	22	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	7.5	11.9	16.7	22.6	25.7	28.5	27.7	25.7	20.4	13.5	9.8
		MCWB	7.6	5.5	7.8	10.4	17.3	19.5	20.1	21.1	22.1	18.0	11.3	8.3
	2%	DB	6.0	5.2	9.3	13.7	19.6	23.6	26.0	25.9	24.3	18.8	12.2	8.0
		MCWB	5.1	3.6	6.5	8.8	15.6	19.1	20.9	21.6	21.0	16.5	10.2	6.8
	5%	DB	4.5	3.7	7.4	12.0	17.9	22.1	24.7	24.7	23.0	17.4	11.0	6.6
		MCWB	3.3	2.1	5.0	8.0	14.2	18.5	20.7	21.4	20.1	15.1	9.2	5.2
	10%	DB	3.0	2.6	5.7	10.5	16.1	20.8	23.7	23.8	21.8	16.0	9.7	5.2
		MCWB	1.9	1.2	3.4	7.3	12.8	17.7	20.2	20.8	19.0	13.6	8.0	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	5.8	9.5	12.3	18.8	21.8	23.6	24.1	23.1	19.2	12.1	8.7
		MCDB	8.2	6.9	10.8	15.3	21.5	23.8	25.5	25.5	24.9	19.7	13.1	9.6
	2%	WB	5.2	3.7	7.1	10.3	16.4	20.3	22.5	23.1	21.8	16.9	10.7	7.0
		MCDB	5.8	4.8	8.4	12.6	18.7	22.3	24.5	24.8	23.7	18.5	11.7	7.9
	5%	WB	3.4	2.4	5.4	8.9	14.9	19.2	21.7	22.3	20.7	15.6	9.5	5.5
		MCDB	4.2	3.6	7.0	11.0	17.2	21.2	23.7	24.1	22.4	17.1	10.7	6.4
	10%	WB	1.9	1.3	3.8	7.7	13.4	18.2	20.9	21.4	19.6	14.1	8.4	4.0
		MCDB	2.9	2.5	5.5	9.9	15.6	20.3	22.9	23.3	21.4	15.6	9.6	5.0
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.1	6.7	6.8	7.0	7.0	6.8	6.9	7.3	6.6	6.2	6.1
		MCDBR	6.2	6.5	7.4	8.9	9.0	8.3	7.7	7.3	7.4	7.3	6.9	6.3
		MCWBR	6.3	6.0	5.8	6.1	5.8	5.0	4.8	5.3	5.7	6.1	6.3	6.0
	5% WB	MCDBR	6.3	6.5	6.9	7.9	8.0	7.2	6.4	6.2	6.6	6.8	6.4	6.0
		MCWBR	6.5	6.3	5.9	6.2	6.0	5.2	4.8	5.0	5.4	6.1	6.3	6.2
Clear-Sky Solar Irradiance	taub	0.306	0.311	0.325	0.363	0.389	0.424	0.421	0.411	0.373	0.358	0.314	0.290	
	taud	2.377	2.366	2.424	2.372	2.349	2.282	2.303	2.320	2.409	2.430	2.506	2.483	
	Ebn at Noon	815	880	916	901	882	849	847	843	851	815	808	809	
	Edh at Noon	78	93	101	115	122	131	127	120	102	88	69	64	
All-Sky Solar Radiation	RadAvg	1.30	2.15	3.46	4.47	5.47	5.85	5.95	5.22	4.16	2.46	1.52	1.05	
	RadStd	0.13	0.19	0.32	0.48	0.57	0.47	0.45	0.27	0.34	0.22	0.19	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page